

Getting Started with GNU Octave

Last month, we learnt how to use Octave's programming language so that we could write standalone Octave scripts, user-defined functions, and the like. This month, we move beyond Octave itself, and learn how to access functions written in C++ from Octave, and vice versa.

Let me give you a couple of reasons to encourage you to learn how to access functions written in C++ from Octave, and vice versa. Let's say you already have a large application built in C++; Octave doesn't have that application's functionality, and you want to avoid rewriting the whole code in Octave's language. Here, you need to access your existing application from Octave.

Next, consider the reverse case—let's assume you just want to call one of the numerical functions of Octave, or make use of the data structures supported by Octave, from your C++ code. This way, you can avoid implementing the same functionality in C++, and just call the Octave function instead.

Both are good cases for interoperability between C++ and Octave. So let's start off with how we can access functions written in C++ from Octave. As you might have guessed, you'll need at least some basic familiarity with C++.

Oct-files

An Oct-file is a dynamic extension to the Octave interpreter; you can also call it a shared library. It is written in C++, including the necessary Octave header files. It is then

compiled using *mkoctfile*, to create shared libraries or standalone C++ programs (our second goal in this article). *Mkoctfile* is an Octave utility (callable from within Octave, or from the command line).

Writing your first Octave extension

Our first Octave extension will be simple—just print “Hello World” in Octave. Consider the code listing below:

```
#include <octave/oct.h>

DEFUN_DLD (hello_octave, args, nargout,
           "I am a Help String")
{
    int nargin = args.length ();
    octave_stdout << "This function has " << nargin
                  << " input arguments and "
                  << nargout << " output arguments.\n";
    return octave_value_list ();
}
```

Save this into the file *hello_octave.cc*.

From the terminal or your Octave prompt, run *mkoctfile hello_octave.cc*. You should now see that you have new files in the folder: *hello_octave.o* and *hello_octave.oct*. Now, let us load